

Day 3 Practice Sheet — NumPy Foundations

Module: NumPy & Pandas — NumPy Foundations

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Objective: Practise array creation, reshaping, slicing, broadcasting, and statistical functions.

INSTRUCTIONS

All questions are answered in a Jupyter Notebook. Import NumPy before starting: `import numpy as np`

PRACTICE QUESTIONS

Question	Task Description	Your Answer / Code
Q1 [Array Creation]	Create a 1D NumPy array of integers from 1 to 20 (inclusive) using <code>np.arange()</code> .	
Q2 [Array Creation]	Create a 4×5 array of zeros. Then create another 4×5 array of ones. Add them together.	
Q3 [Array Creation]	Create an array of 8 evenly spaced values between 0 and 1 using <code>np.linspace()</code> . Print the values.	
Q4 [Reshaping]	Create <code>np.arange(1, 25)</code> and reshape it into a 4×6 array. Print the shape, number of dimensions (<code>ndim</code>), and data type (<code>dtype</code>).	
Q5 [Slicing 1D]	From the array <code>[10, 20, 30, 40, 50, 60, 70, 80, 90, 100]</code> , extract: (a) first 5 elements, (b) last 3 elements, (c) every alternate element.	
Q6 [Slicing 2D]	Create a 5×5 array using <code>np.arange(25).reshape(5,5)</code> . Extract: (a) row 2, (b) column 3, (c) top-right 3×3 block.	

Q7 [Broadcasting]	Create array <code>a = [1, 2, 3, 4, 5]</code> . Add 100 to every element. Multiply every element by 3. Both without any loop.	
Q8 [Broadcasting]	Create two arrays: <code>a = [10, 20, 30]</code> and <code>b = [1, 2, 3]</code> . Compute: <code>a+b</code> , <code>a-b</code> , <code>a*b</code> , <code>a/b</code> element-wise.	
Q9 [Statistical Ops]	Given scores = <code>[72, 85, 91, 68, 74, 88, 62, 95, 77, 80]</code> , compute: mean, median (<code>np.median</code>), std, min, max, and range (<code>max-min</code>).	
Q10 [Statistical Ops]	For a 3x4 random integer array (<code>np.random.randint(1,100,size=(3,4))</code>): compute sum across all elements, row-wise sum (<code>axis=1</code>), and column-wise mean (<code>axis=0</code>).	
BONUS	Generate 1000 random numbers from a normal distribution (<code>np.random.normal(mean=50, std=10, size=1000)</code>). Compute mean and std. Are they close to 50 and 10?	